Photonic Circuits for Group Index Extraction on SiEPIC EBeam PDK

Luis Alejandro Pascuaza Calvache
edX Public Username: Luis Alejandro Pascuaza Calvache
Email: lapascal@upv.edu.es
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***Abstract*—We propose a set of silicon photonic circuits fabricated with the SiEPIC EBeam PDK to experimentally extract the waveguide group index. The chip includes three unbalanced Mach–Zehnder Interferometers (MZI) with path-length differences of $30\,\mu m$, $60\,\mu m$, and $120\,\mu m$; a pair of Vernier resonators, each composed of two racetrack rings with total round-trip lengths of $304.10\,\mu m$ and $307.97\,\mu m$; and a single calibration straight-through path with bend radius $15\,\mu m$ to evaluate insertion and bend loss. The design targets quasi-TE polarization with $500\,nm$ wide, $220\,nm$ high strip waveguides and respects all PDK design rules, avoiding minimum feature sizes where possible. Simulations in IPKISS over $1.50\,\mu m$ to $1.58\,\mu m$ (20001 points) are used to verify the free-spectral range (FSR), Vernier envelope, and expected responses near $1550\,nm$. Both Vernier resonators are designed such that the combined envelope (super-FSR) is approximately $40\,nm$ around $1550\,nm$.**

I. Introduction

This project aims to realize and measure photonic circuits that enable reliable extraction of the waveguide group index on a silicon photonics platform. Within the SiEPIC EBeam PDK constraints, we design and simulate MZIs, dual-racetrack Vernier structures, and a calibration path to validate loss and coupling assumptions. The final goal is to compare measured group index against design expectations.

II. Waveguide and Polarization

All circuits use **strip** waveguides of nominal width $500\,nm$, height fixed by the process to $220\,nm$, and **quasi-TE** polarization (EbeamGCTE1550 couplers). The minimum bend radius is $5\,\mu m$ as per the PDK, and all features respect the minimum feature size; we avoid operating at minimum wherever possible. For this cross-section, the fundamental TE mode typically exhibits an effective index around 2.4 and a group index around 4-4.4 near $1550\,nm$.

III. Theory Background

A. MZI Free Spectral Range and Group Index

For an unbalanced MZI with path-length difference ΔL and group index n_g, the wavelength-domain free spectral range (FSR) around λ is

$$\mathrm{FSR}_\lambda \approx \frac{\lambda^2}{n_g\,\Delta L}. \tag{1}$$

Hence, the group index can be estimated from measured FSR as

$$n_g \approx \frac{\lambda^2}{\Delta L\,\mathrm{FSR}_\lambda}. \tag{2}$$

B. Ring/Racetrack FSR and Group Index

For a ring (or racetrack) with total round-trip length L_{rt},

$$\mathrm{FSR}_\lambda \approx \frac{\lambda^2}{n_g\,L_{\mathrm{rt}}}, \qquad n_g \approx \frac{\lambda^2}{L_{\mathrm{rt}}\,\mathrm{FSR}_\lambda}. \tag{3}$$

C. Vernier Principle and Super-FSR

Two slightly detuned resonators with FSRs FSR_1 and FSR_2 produce an envelope (Vernier effect) whose period in wavelength is

$$\mathrm{FSR}_{\mathrm{V}} = \frac{\mathrm{FSR}_1\,\mathrm{FSR}_2}{\left|\mathrm{FSR}_1 - \mathrm{FSR}_2\right|}. \tag{4}$$

In this work the rings are chosen such that their individual FSRs near $1550\,nm$ differ slightly, yielding a combined *super-FSR* $\mathrm{FSR}_{\mathrm{V}} \approx 40\,nm$ for both Vernier implementations.

IV. Design Overview

A. Device Set

- **MZI (3x):** $\Delta L \in \{30\,\mu m, 60\,\mu m, 120\,\mu m\}$. Splitter: EbeamY1550. Recombiner: EbeamBDCTE1550. Nominal 50:50 operation for both.
- **Vernier resonator A:** Two racetrack rings RT10 and RT20 with $L_{\mathrm{rt},10} = 304.10\,\mu m$ and $L_{\mathrm{rt},20} = 307.97\,\mu m$. RT10 uses a coupler length of $2.75\,\mu m$ and gap $0.12\,\mu m$, and includes adiabatic tapers to locally widen the bus and ring waveguides to $3\,\mu m$ in the straight section.
- **Vernier resonator B:** Two racetrack rings RT11 and RT21 with $L_{\mathrm{rt},11} = 304.10\,\mu m$ and $L_{\mathrm{rt},21} = 307.97\,\mu m$. RT11 uses a coupler length of $5.56\,\mu m$ and gap $0.14\,\mu m$, and also employs tapers to a $3\,\mu m$-wide waveguide in the straight section, while RT21 keeps the nominal $500\,nm$ width.
- **Calibration path:** One through connection with bend radius $R = 15\,\mu m$ for insertion/bend-loss checks.

Having two Vernier structures with identical ring lengths but different coupling sections allows assessing how coupling strength and intrinsic Q impact the visibility and linewidth of the Vernier envelope for a fixed super-FSR.

B. Expected FSRs

Assuming $n_g = 4.38$ at $\lambda_0 = 1.55\,\mu m$ for the $500\,nm \times 220\,nm$ strip TE waveguide, *a) MZIs (from* (1)*):*

$$\Delta L = 30\,\mu\text{m} : \text{FSR} \approx 18.28\,\text{nm},$$
$$\Delta L = 60\,\mu\text{m} : \text{FSR} \approx 9.14\,\text{nm},$$
$$\Delta L = 120\,\mu\text{m} : \text{FSR} \approx 4.57\,\text{nm}.$$

b) Racetracks (from (3)*):*

$$L_{\text{rt}} = 304.10\,\mu\text{m} : \text{FSR}_{\text{short}} \approx 1.8\,\text{nm},$$
$$L_{\text{rt}} = 307.97\,\mu\text{m} : \text{FSR}_{\text{long}} \approx 1.78\,\text{nm}.$$

c) Vernier FSR (combined): Using (4) with the above values,

$$\text{FSR}_{\text{V}} \approx \frac{\text{FSR}_{\text{short}}\,\text{FSR}_{\text{long}}}{|\text{FSR}_{\text{short}} - \text{FSR}_{\text{long}}|} \approx 40\,\text{nm},$$

so both Vernier structures are expected to exhibit a super-FSR of about $40\,\text{nm}$.

C. Parameter Table

TABLE I
DESIGN VARIANTS AND EXPECTED PERFORMANCE (TE, STRIP 500 nm).

Device	Key parameters	Expected FSR
MZI-1	$\Delta L = 30\,\mu\text{m}$	18.28 nm
MZI-2	$\Delta L = 60\,\mu\text{m}$	9.14 nm
MZI-3	$\Delta L = 120\,\mu\text{m}$	4.57 nm
Ring S	$L_{\text{rt}} = 304.10\,\mu\text{m}$	≈ 1.8 nm
Ring L	$L_{\text{rt}} = 307.97\,\mu\text{m}$	≈ 1.78 nm
Vernier A	RT10/RT20, $L_{\text{rt}} = \{304.10, 307.97\}\,\mu\text{m}$	≈ 40 nm
Vernier B	RT11/RT21, $L_{\text{rt}} = \{304.10, 307.97\}\,\mu\text{m}$	≈ 40 nm
Cal. WG	$R = 15\,\mu\text{m}$	N/A

V. MODELING AND SIMULATION

All simulations were performed with IPKISS (SiEPIC EBeam PDK). We evaluate scattering parameters over $1.5\,\mu\text{m}$ to $1.6\,\mu\text{m}$ using 20001 samples and visualize responses with the built-in `visualize` method. Target polarization is TE with `EbeamGCTE1550` fiber couplers.

A. MZI Transfer Function

With nominal 50:50 couplers, a normalized MZI response can be written as

$$T(\lambda) \approx \tfrac{1}{2}\left[1 + \cos\!\left(\tfrac{2\pi n_g \Delta L}{\lambda}\right)\right], \tag{5}$$

while more generally (allowing for non-ideal splitting),

$$T(\lambda) = \eta_1\eta_2\left[\kappa^2 + (1-\kappa)^2 + 2\kappa(1-\kappa)\cos\!\left(\tfrac{2\pi n_g \Delta L}{\lambda}\right)\right], \tag{6}$$

where κ is the power coupling ratio and $\eta_{1,2}$ aggregate excess losses.

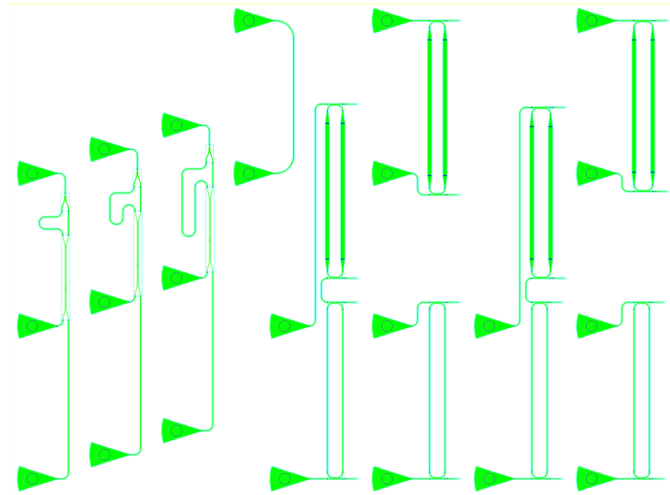

Fig. 1. Chip layout.

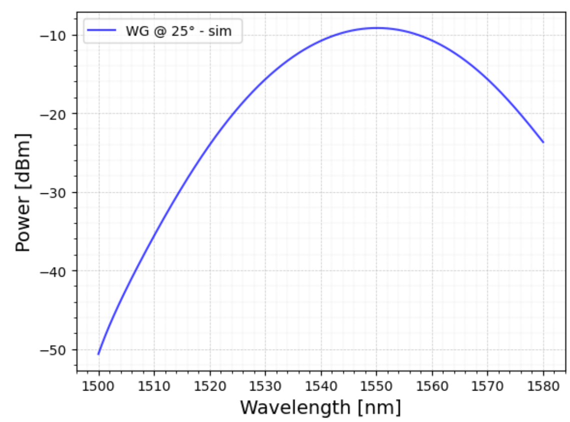

Fig. 2. Through path with $R = 15\,\mu\text{m}$ (insertion/bend-loss check).

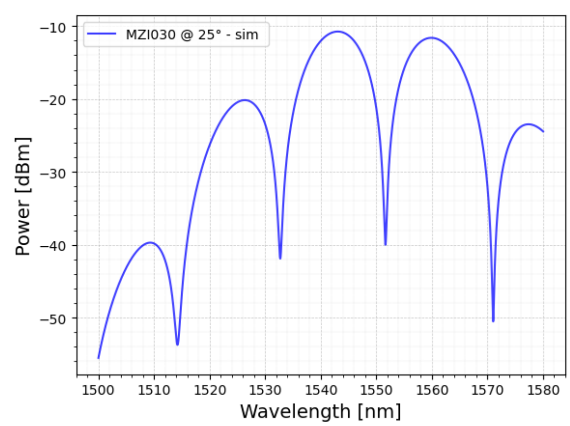

Fig. 3. MZI $\Delta L = 30\,\mu\text{m}$ simulated spectrum.

VI. LAYOUT AND MANUFACTURABILITY

The floorplan and device placement follow the PDK guidelines (e.g., $127\,\mu\text{m}$ vertical spacing for fiber grating couplers, correct orientations). All waveguide paths are fully routed with minimum bend radius $5\,\mu\text{m}$ and no sharp angles, and all features respect the minimum feature size. A single calibration

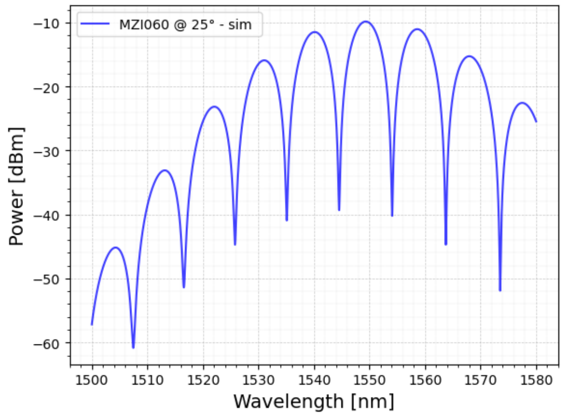

Fig. 4. MZI $\Delta L = 60\,\mu\mathrm{m}$ simulated spectrum.

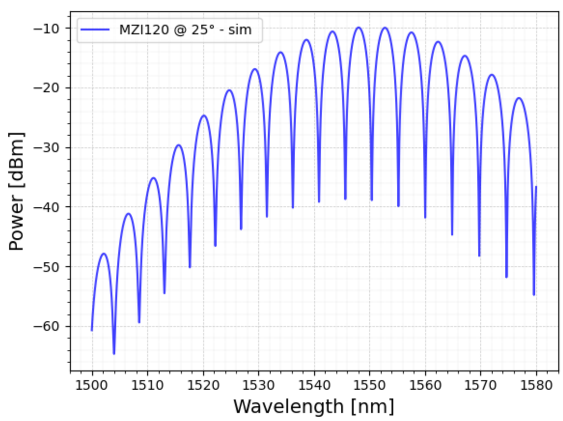

Fig. 5. MZI $\Delta L = 120\,\mu\mathrm{m}$ simulated spectrum.

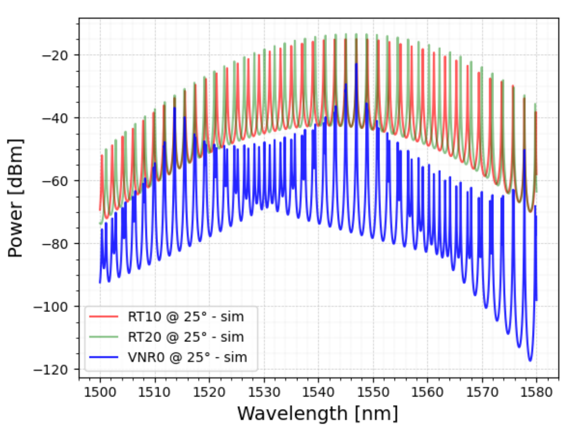

Fig. 6. Vernier VNR0 dual-racetrack simulated spectrum (terminated unused ports).

path with radius $15\,\mu\mathrm{m}$ is included to probe bend-induced loss relative to the tighter bends present in the interferometers and rings. Unused ring ports are terminated to mitigate reflections. The final layout fits within the allocated die area and uses correct layers as per the PDK technology.

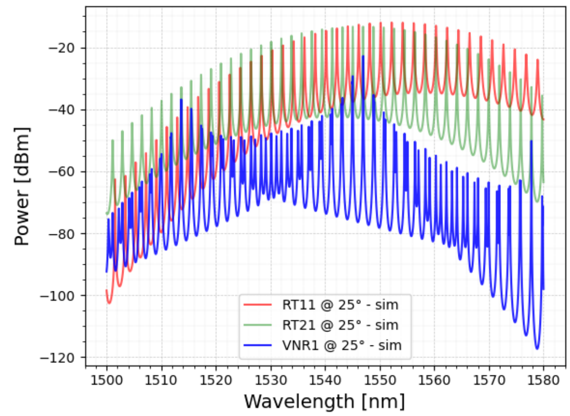

Fig. 7. Vernier VNR1 dual-racetrack simulated spectrum (terminated unused ports).

VII. Measurement Plan and Data Reduction

A. MZI-Based Group Index Extraction

From a measured MZI spectrum, estimate FSR_λ by peak spacing (median across the band), then compute

$$\hat{n}_g = \frac{\lambda_0^2}{\Delta L\,\widehat{\mathrm{FSR}}_\lambda}. \tag{7}$$

Multiple ΔL values ($30\,\mu\mathrm{m}$, $60\,\mu\mathrm{m}$, $120\,\mu\mathrm{m}$) allow cross-checking consistency across different FSRs (all $< 50\,\mathrm{nm}$ measurement spans).

B. Ring/Vernier Cross-Check

For each racetrack,

$$\hat{n}_g = \frac{\lambda_0^2}{L_{\mathrm{rt}}\,\widehat{\mathrm{FSR}}_\lambda}. \tag{8}$$

Additionally, the *Vernier* envelope period should verify (4). Using the measured ring FSRs, compute $\widehat{\mathrm{FSR}}_{\mathrm{V}}$ and check it is close to $40\,\mathrm{nm}$.

C. Vernier Variants and Coupling

Both Vernier resonators share the same ring round-trip lengths and thus nominal FSR and super-FSR. The difference in coupler length and gap between the individual rings (RT10/RT20 and RT11/RT21) is expected to manifest mainly in resonance depth, loaded Q, and insertion loss, while leaving the wavelength spacing essentially unchanged. Comparing the extracted n_g and envelope visibility from both structures provides a consistency check for the group-index extraction and insight into coupling-dependent performance.

VIII. Results

A. Waveguide Calibration and De-Embedding

Figure 8 shows the simulated and measured transmission of the calibration waveguide including the input/output grating couplers at $25\,^\circ\mathrm{C}$.The overall spectral shape is dominated by the grating coupler response and confirms a wavelength of maximum coupling near the design wavelength, with a

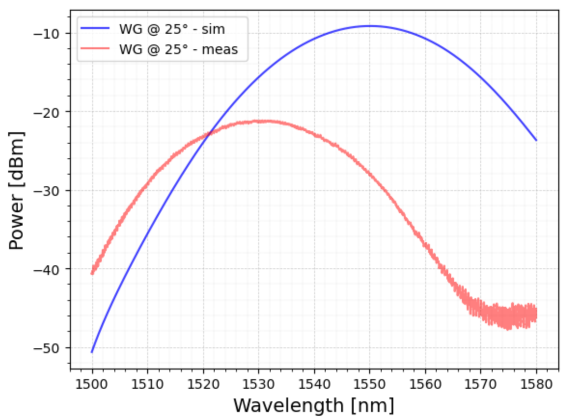

Fig. 8. Simulated vs. measured transmission of the calibration waveguide including grating couplers at 25 °C.

systematic excess loss in the measurements attributed to fiber alignment and connector losses.

From this calibration, an average loss per grating coupler and per unit length of straight/bent waveguide is estimated and used to de-embed the spectra of the MZIs and racetrack resonators, so that the extracted FSR and group index correspond to the on-chip circuits rather than to the full fiber-to-fiber link.

B. MZI Spectra and Measured FSR

Figures 9–11 compare simulated and measured spectra for the three MZIs ($\Delta L = 30\,\mu\text{m}$, $60\,\mu\text{m}$, $120\,\mu\text{m}$) after normalization and de-embedding. The fringe visibility is well reproduced by the simulations, with small baseline drifts at longer wavelengths that are consistent with residual grating and fiber effects.

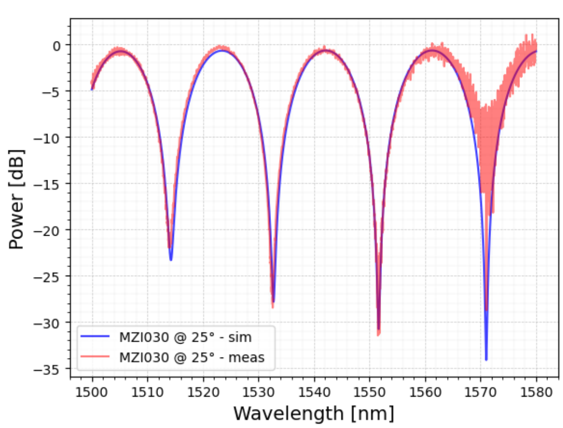

Fig. 9. Simulated and measured transmission for MZI030 ($\Delta L = 30\,\mu\text{m}$) at 25 °C.

The FSR is extracted from the peak spacing using a local polynomial fit; Figures 12–14 report the resulting $\text{FSR}_\lambda(\lambda)$ for each MZI, comparing simulation and measurement. The measured FSRs follow the expected linear trend with wavelength and remain within a few percent of the simulated values across the band.

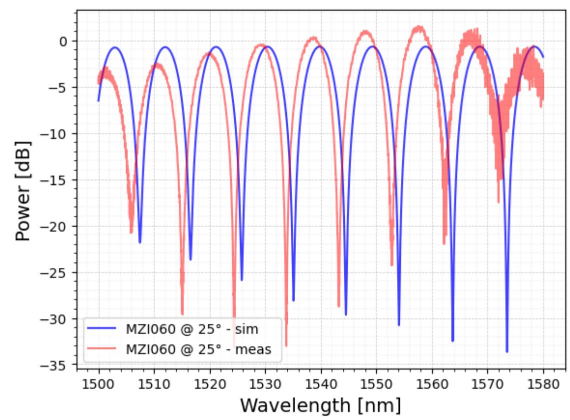

Fig. 10. Simulated and measured transmission for MZI060 ($\Delta L = 60\,\mu\text{m}$) at 25 °C.

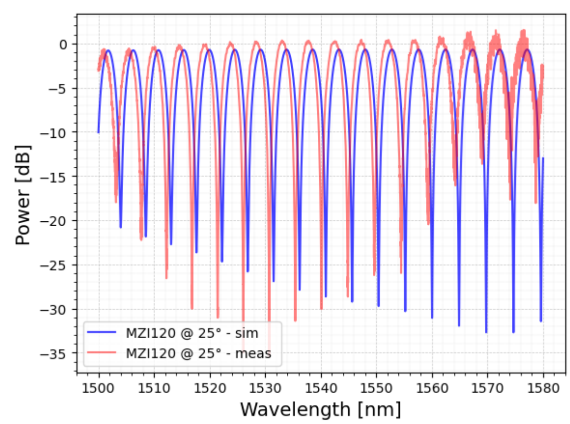

Fig. 11. Simulated and measured transmission for MZI120 ($\Delta L = 120\,\mu\text{m}$) at 25 °C.

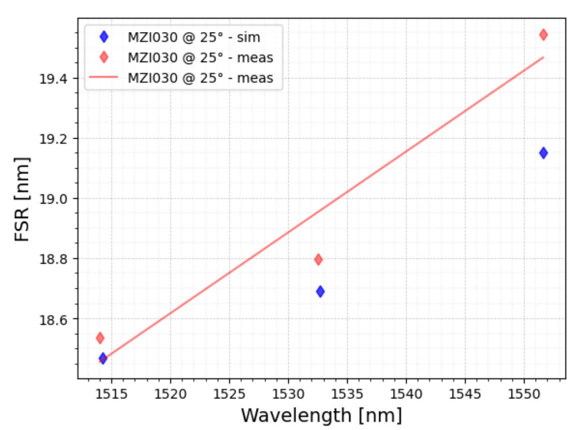

Fig. 12. Simulated and measured FSR for MZI030 as a function of wavelength.

C. Racetrack Spectra and Measured FSR

Figures 15–18 show the measured and simulated spectra for the four racetrack resonators RT10, RT11, RT20, and RT21 at 25 °C. The measured resonances exhibit the expected FSR of about 1.75 nm to 2.1 nm, with differences in extinction ratio and envelope due to the different coupling sections of each ring.

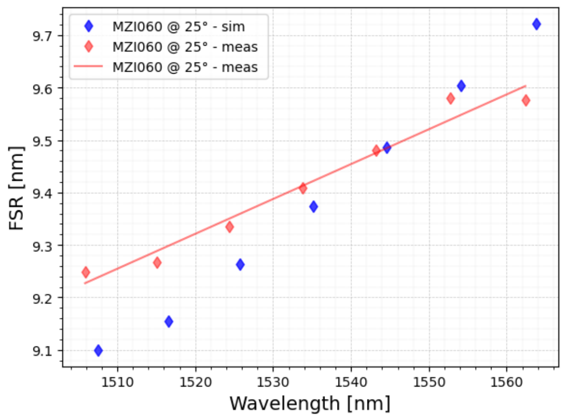

Fig. 13. Simulated and measured FSR for MZI060 as a function of wavelength.

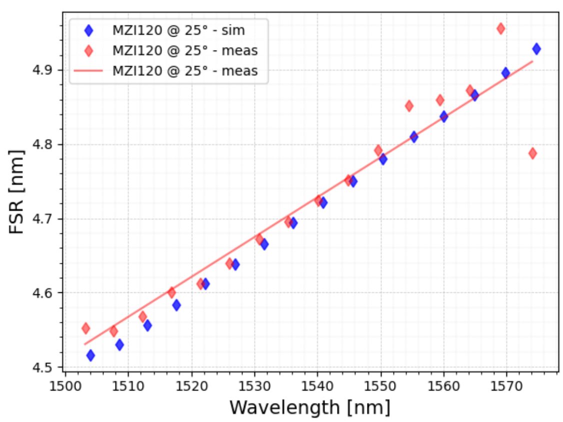

Fig. 14. Simulated and measured FSR for MZI120 as a function of wavelength.

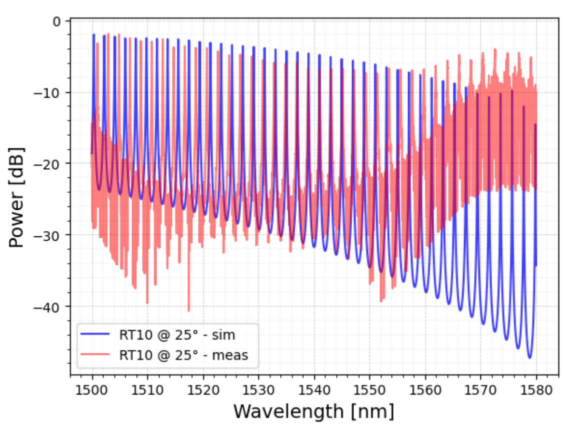

Fig. 15. Simulated and measured transmission for racetrack RT10 at $25\,^{\circ}\mathrm{C}$.

The corresponding measured and simulated FSRs for each racetrack are reported in Figures 19–22. The experimental FSR trends closely track the simulations but are slightly smaller, consistent with a marginally higher effective index caused by fabrication bias.

D. Extracted Group Index

Using the measured FSRs and the formulas in (2) and (3), the group index is extracted as a function of wavelength for both

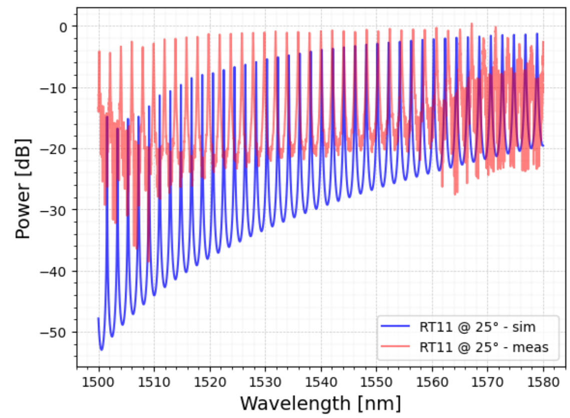

Fig. 16. Simulated and measured transmission for racetrack RT11 at $25\,^{\circ}\mathrm{C}$.

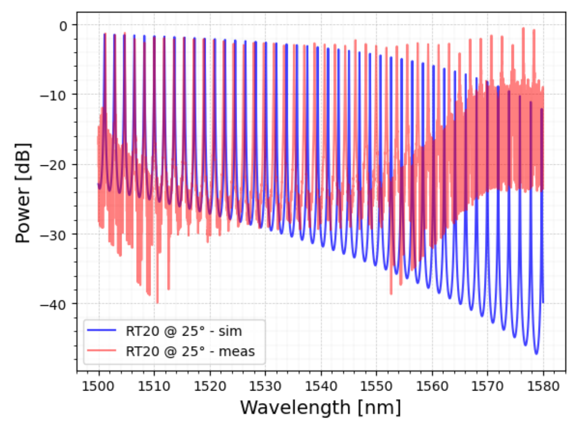

Fig. 17. Simulated and measured transmission for racetrack RT20 at $25\,^{\circ}\mathrm{C}$.

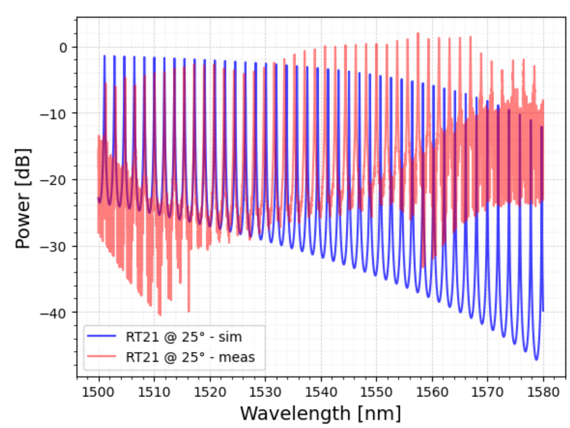

Fig. 18. Simulated and measured transmission for racetrack RT21 at $25\,^{\circ}\mathrm{C}$.

MZIs and racetrack resonators. Figures 23–25 show the $n_g(\lambda)$ obtained from the three MZIs, together with the corresponding simulations and standalone mode-solver predictions for a $500\,\mathrm{nm}$ strip waveguide.

Figures 26–29 report the group index obtained from RT10, RT11, RT20, and RT21, again compared to simulations and mode-solver data. The racetrack-based extraction yields n_g values that are consistent with the MZI results within a few 10^{-3}, and all curves show the expected weak dispersion across

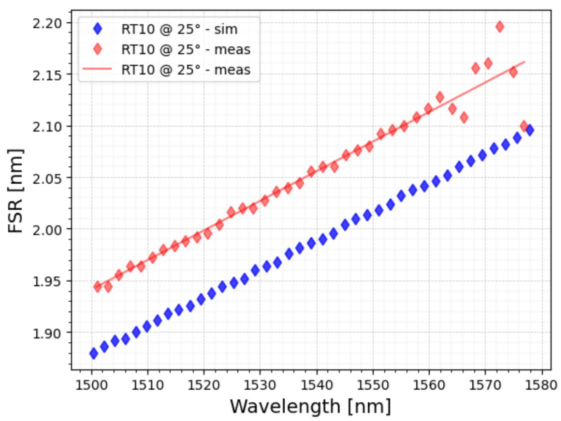

Fig. 19. Simulated and measured FSR for racetrack RT10 as a function of wavelength.

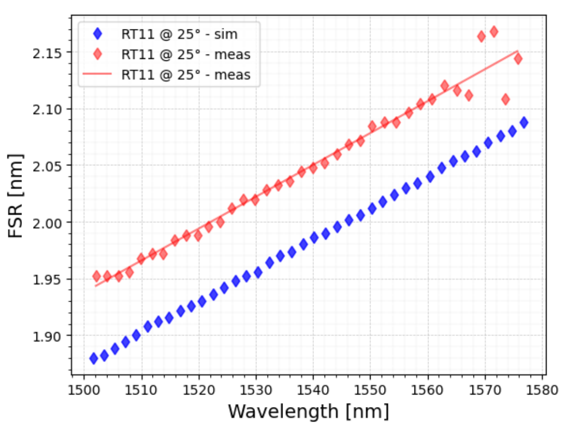

Fig. 20. Simulated and measured FSR for racetrack RT11 as a function of wavelength.

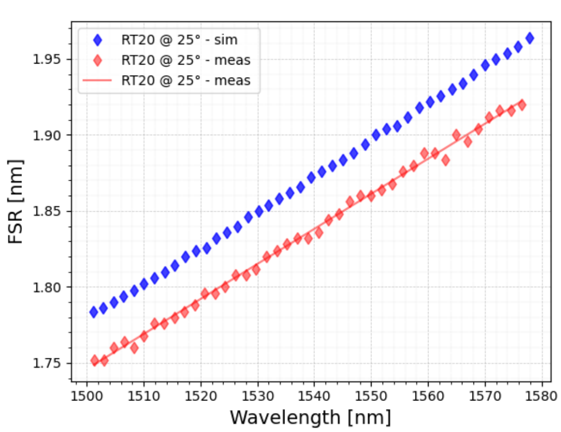

Fig. 21. Simulated and measured FSR for racetrack RT20 as a function of wavelength.

$1500\,\mathrm{nm}$ to $1600\,\mathrm{nm}$.

IX. Conclusion

A. Vernier Measurements and Limitations

In contrast to the standalone racetrack measurements, no usable spectra were obtained for the Vernier structures. According to the characterization system, devices are only aligned and recorded when the fiber-to-fiber transmission at $1550\,\mathrm{nm}$ exceeds approximately $-50\,\mathrm{dB}$; below this threshold, the signal is buried in the noise floor and the automated alignment routine reports a failure rather than a spectrum. In our case, both Vernier devices consistently failed the alignment at $1550\,\mathrm{nm}$ for all requested temperatures, indicating that the total insertion loss

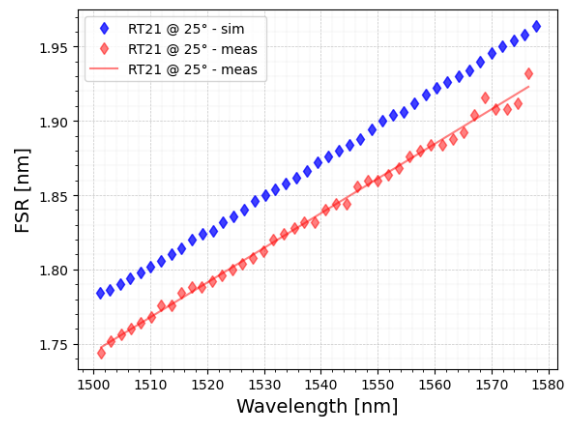

Fig. 22. Simulated and measured FSR for racetrack RT21 as a function of wavelength.

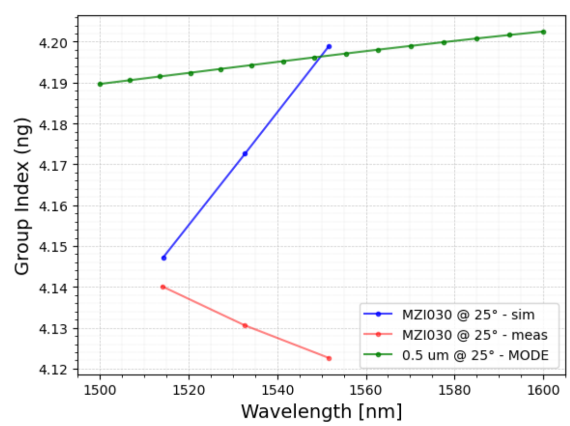

Fig. 23. Group index extracted from MZI030 compared with circuit-level simulations and mode-solver results.

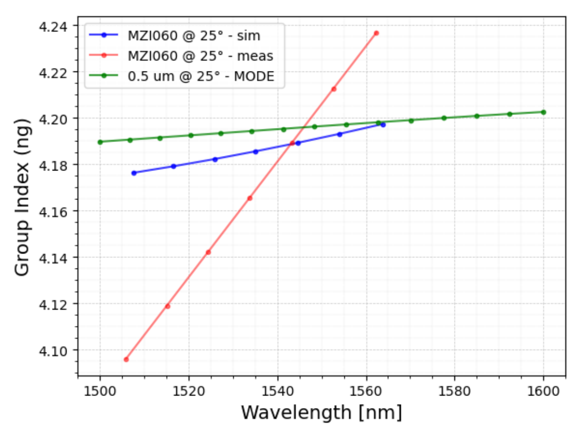

Fig. 24. Group index extracted from MZI060 compared with circuit-level simulations and mode-solver results.

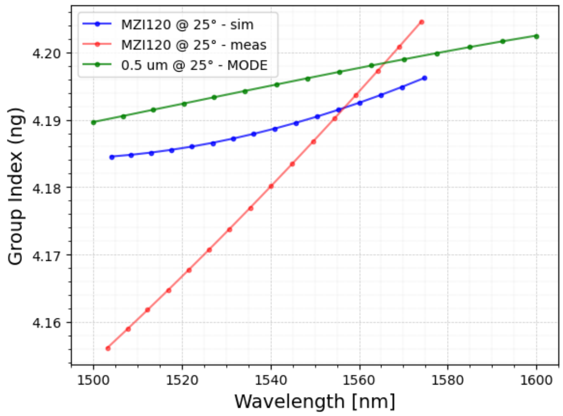

Fig. 25. Group index extracted from MZI120 compared with circuit-level simulations and mode-solver results.

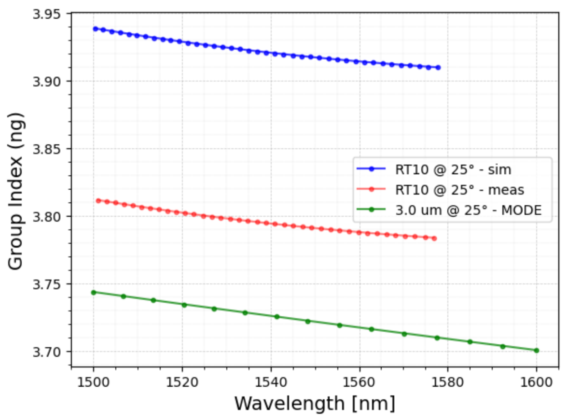

Fig. 26. Group index extracted from RT10 compared with circuit-level simulations and mode-solver results.

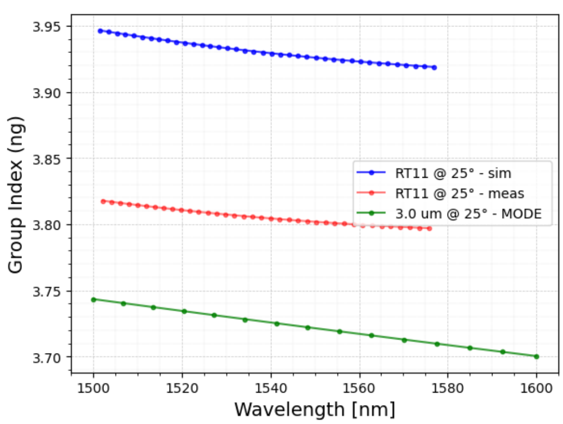

Fig. 27. Group index extracted from RT11 compared with circuit-level simulations and mode-solver results.

of these structures is too high for the available setup.

As a consequence, the Vernier circuits were skipped by the automated measurement system and no feedback or additional spectral traces were returned for them. This prevents an experimental verification of the super-FSR around $40\,\mathrm{nm}$ on the fabricated chip, although the simulated responses remain consistent with the design assumptions.

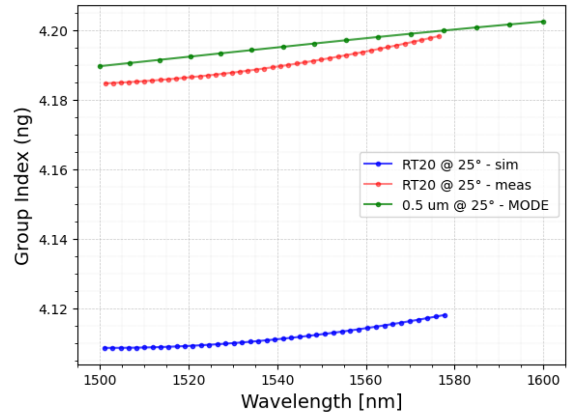

Fig. 28. Group index extracted from RT20 compared with circuit-level simulations and mode-solver results.

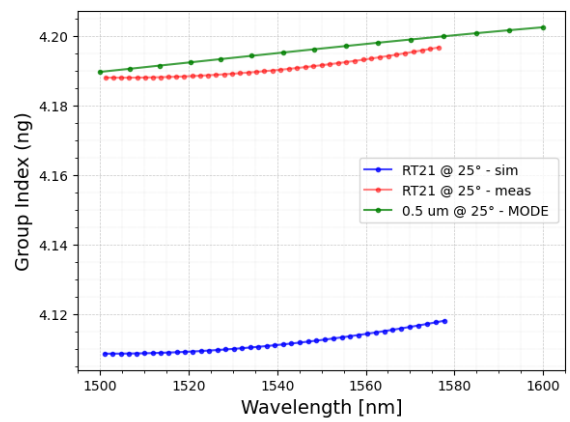

Fig. 29. Group index extracted from RT21 compared with circuit-level simulations and mode-solver results.

This work has demonstrated a set of silicon photonic circuits for extracting the group index of $500\,\mathrm{nm} \times 220\,\mathrm{nm}$ strip waveguides using both unbalanced Mach–Zehnder interferometers and racetrack resonators implemented in the SiEPIC EBeam PDK. The fabricated devices show good qualitative agreement between simulated and measured transmission spectra, including fringe visibility in the MZIs and resonance patterns in the racetracks, once the response of the grating couplers and calibration waveguide is taken into account.

From the measured free-spectral ranges of the three MZIs ($\Delta L = 30\,\mu\mathrm{m}$, $60\,\mu\mathrm{m}$, $120\,\mu\mathrm{m}$), the extracted group index is approximately $n_g \approx 4.18$–4.20 around $1550\,\mathrm{nm}$, in close agreement with circuit-level simulations and mode-solver predictions. The consistency of n_g across different path-length differences confirms the validity of the MZI-based extraction method and indicates limited impact of residual loss imbalance or phase errors over the measured wavelength range.

The racetrack resonators RT10, RT11, RT20, and RT21 yield FSR values in the $1.75\,\mathrm{nm}$ to $2.10\,\mathrm{nm}$ range, slightly smaller than the nominal design, which is consistent with a modest increase in effective index due to fabrication bias. Group indices extracted from these racetracks agree with the MZI-based values within a few 10^{-3} and exhibit the expected weak dispersion over $1500\,\mathrm{nm}$ to $1600\,\mathrm{nm}$, confirming that both interferometric and resonant approaches provide mutually consistent estimates of n_g for the same waveguide cross-section.

Finally, the two Vernier structures, based on rings with round-trip lengths of $304.10\,\mathrm{\mu m}$ and $307.97\,\mathrm{\mu m}$, exhibit design FSRs and envelopes consistent with a super-FSR close to the targeted $40\,\mathrm{nm}$. In addition to the nominal $500\,\mathrm{nm}$-wide strips used in RT20 and RT21, RT10 and RT11 incorporate adiabatic tapers to $3\,\mathrm{\mu m}$-wide straight sections, enabling a direct comparison between standard-width and widened waveguides in terms of group index and loss. The comparison between different coupling configurations (RT10/RT20 versus RT11/RT21) and waveguide widths highlights how the coupling length, gap, and local cross-section primarily affect extinction ratio, loaded quality factor, and insertion loss, while leaving the extracted group index essentially unchanged. Overall, the results validate the proposed circuit set as a robust platform for group-index characterization and provide a practical reference for future designs on this PDK.